Vegetation Cover in Padova

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1 Introduction & Background

This report investigates the vegetation cover across three distinct functional districts in the city of Padova: the city center, the university area and the industrial area. The study aims to analyze and quantify the green surface area within these zones by calculating a representative average for each district based on multiple samples.

The primary objective of this analysis is to translate visual satellite data into quantitative biological metrics. Specifically, this report seeks to answer:

- What is the average percentage of vegetation cover in the City Center, University Area and Industrial Area of Padova?
- How different urban land uses influence the presence of vegetation?
- Which of these districts exhibits the highest ecological footprint in terms of surface-level vegetation?

1.1 Image Acquisition

For each district, five distinct images were acquired to ensure a representative average of the urban greenery. Given Padova's character as a university city with academic facilities integrated into the urban fabric, the University district was delimited to the Portello and Piovego areas, where the primary concentration of educational hubs is located.

To conduct this analysis, the data were collected using satellite and aerial remote sensing via **Google Earth Pro**; the images in Google Earth are taken by cameras on satellites and aircraft, which collect each image at a specific date and time.

To ensure the accuracy of the area calculations and the reproducibility of the dataset, several rules were followed:

- **Projection and Orientation:** Images were acquired as 2D projections. The "3D Buildings" layer was deactivated to eliminate perspective distortion, ensuring that vertical structures did not obscure the ground-level vegetation. Before each capture, the view was reset to ensure a perfectly perpendicular orientation with North at the top.
- **Scale and Resolution Consistency:** A constant Eye Altitude of 1150 feet was maintained, providing a uniform spatial resolution across all samples, and further calibrated by a 300-foot scale bar included in every image.

- **Temporal Standardization:** All images were gathered from June 2025. Selecting the summer season is biologically significant, since the vegetation is at its peak and the area of the trees are the maximum, allowing for a more comprehensive pixel-based quantification of the tree canopy.
- **Systematic Sampling:** To eliminate observer bias, a grid-based sampling method was used; this ensured that image placement was determined by spatial coordinates rather than a subjective preference for greener or more interesting patches.
- **Overlap Prevention:** After capturing each sample, a digital placemark was dropped on the map to serve as a spatial reference point and to ensure that the images didn't overlap between samples.
- **Naming Convention:** To ensure traceability, a standardized nomenclature was applied to the entire dataset using the format: `pd_{district}_{image_number}`.

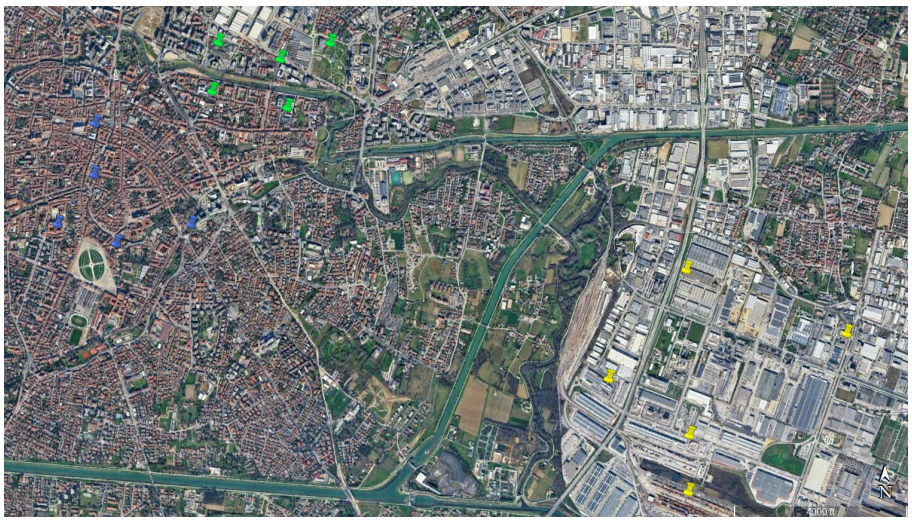


Figure 1: Padova map with placemark where the samples were taken.

2 Data Analysis

The quantitative analysis of vegetation cover was performed using a two-step computational pipeline: first **Ilastik** was used to execute the image segmentation, followed by an automated pixel quantification on **ImageJ**.

The combination of Ilastik and ImageJ was chosen because Ilastik's Random Forest classifier is far more effective at handling complex urban textures (like green-colored roofs or shadows) than simple color thresholding. ImageJ's scripting capabilities then provided the necessary speed and mathematical accuracy to handle the statistics of a large dataset without manual intervention.

2.1 Ilastik: Pixel Classification

The aim of this step was to distinguish between biological and anthropic surfaces using Ilastik, an interactive learning and segmentation software.

The features were selected based on Color/Intensity, Edge, and Texture. A Sigma scale ranging from 1.6 to 5.0 was applied in order to allow the software to discriminate between the different classes of pixels and recognize, not only the green color of the trees, but also the specific grainy texture of canopies versus the smooth texture of rooftops.

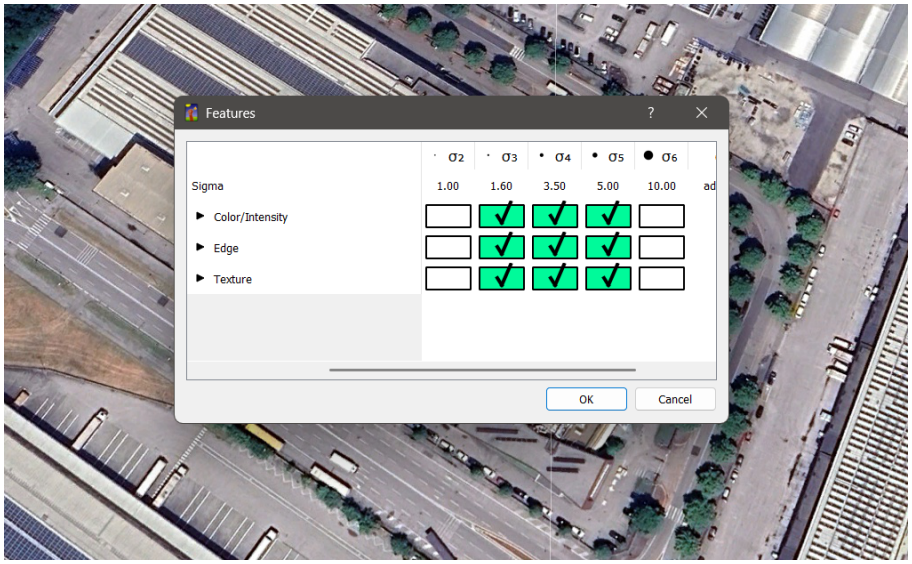


Figure 2: Feature selection dialog for Ilastik analysis.

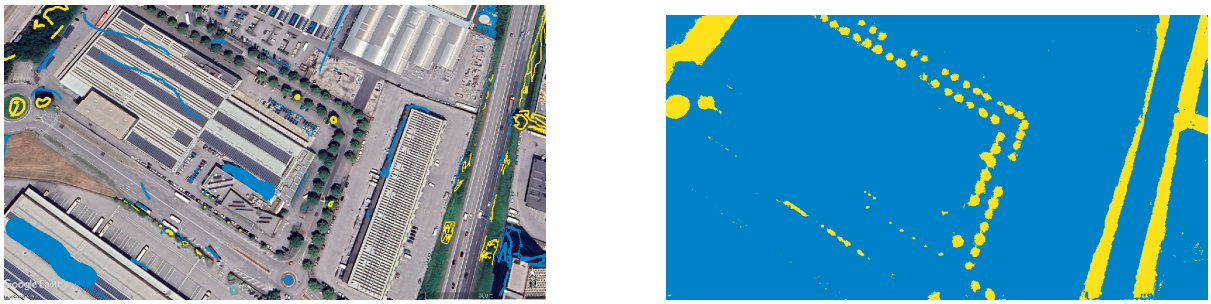
For the labeling step were defined two different classes:

1. **Vegetation:** This class includes tree canopies, grasslands, and shrubs. Additionally, canals and rivers (such as the Piovego) were labeled within this category, as the study aimed to contrast all natural environmental features against the urban fabric.

2. **Anthropic:** This class includes all artificial surfaces, such as rooftops, asphalt roads, concrete structures and vehicles.

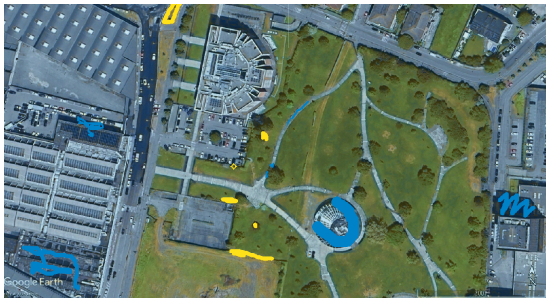
Initial training was performed on a representative image, based on pixel features and user annotations. The remaining images in the dataset were used as test sets; where the classifier struggled with shadows or dark rooftops, additional annotations were drawn to refine the output.

The images were exported as "simple segmentation", creating a binary map in `png` format.

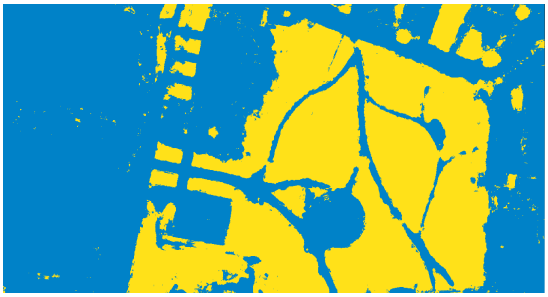


(a) Labeled Image (b) Segmented image

Figure 3: Sample in industrial area.



(a) Labeled Image



(b) Segmented image

Figure 4: Sample in university area.



(a) Labeled Image



(b) Segmented image

Figure 5: Sample in city center.

2.2 ImageJ: Quantitative Image Processing

Once the images were exported as binary segmentation maps, they were processed in ImageJ to extract numerical data.

The segmented images contained discrete pixel values: 1 for Vegetation and 2 for Anthropic. By analyzing the pixel histogram in each image, we could precisely count the frequency of each value.

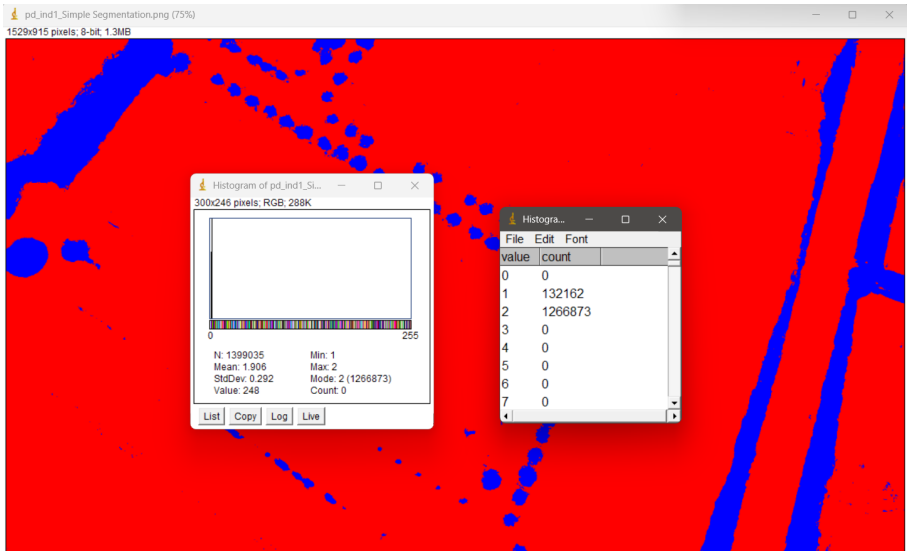


Figure 6: ImageJ histogram values and counting of pixels.

To accelerate the process was developed an ImageJ macro plugin. The script automated the following steps for the entire dataset: after opening each segmented image, it analyzes the pixel distribution by counting the number of pixels with value 1 or 2 and printing the results for each image in a CSV format. At the end, it exports the raw data into a consolidated CSV table for statistical comparison.

```
print("Image, Pixel_Vegetation, Pixel_Anthropic, Total");
for (i=1; i<=nImages; i++) {
    selectImage(i);
    title = getTitle();
    getStatistics(area, mean, min, max, std, histogram);
    vegetation = histogram[1];
    anthropic = histogram[2];
    total = area;
    print(title + ", " + vegetation + ", " + anthropic + ", " + total);
}
```

Listing 1: ImageJ Macro for automated pixel counting

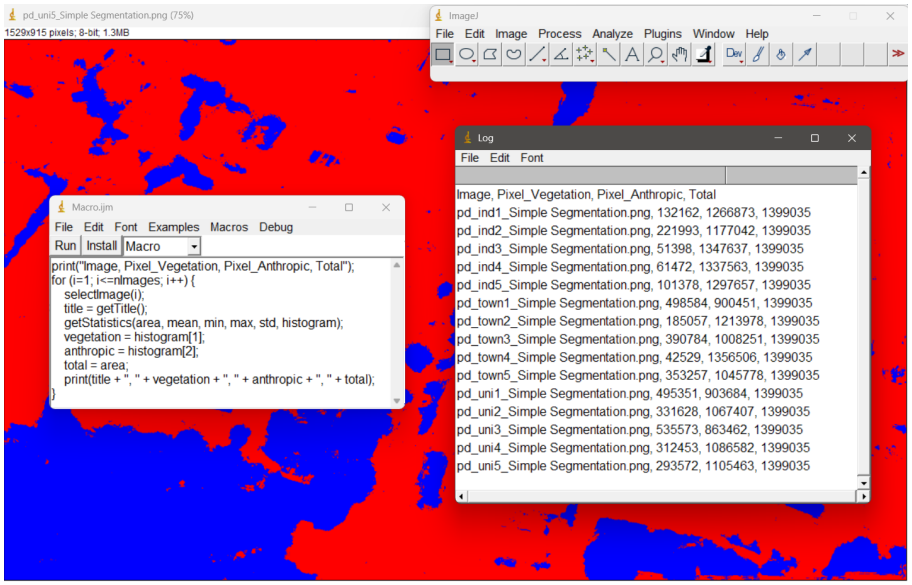


Figure 7: ImageJ Macro display.

3 Presentation of Results

The quantitative analysis of the image dataset revealed significant differences in vegetation density across the three studied districts of Padova; in the following results are summarized the mean percentages of green cover calculated from five independent samples per area.

The data were processed using Python, in specific were used the Pandas library for data manipulation and Matplotlib library for data visualization. The table below displays the calculated mean percentage of vegetation and the associated standard deviation for each district. The bar chart shows that the University Area exhibits the highest biological density, while the Industrial Area shows the lowest, with a significant difference between them.

Table 1: Mean Vegetation Percentage by District

District	Mean Vegetation (%)	Std. Deviation
University Area	28.14%	8.07
City Center	21.02%	13.28
Industrial Area	8.13%	5.3

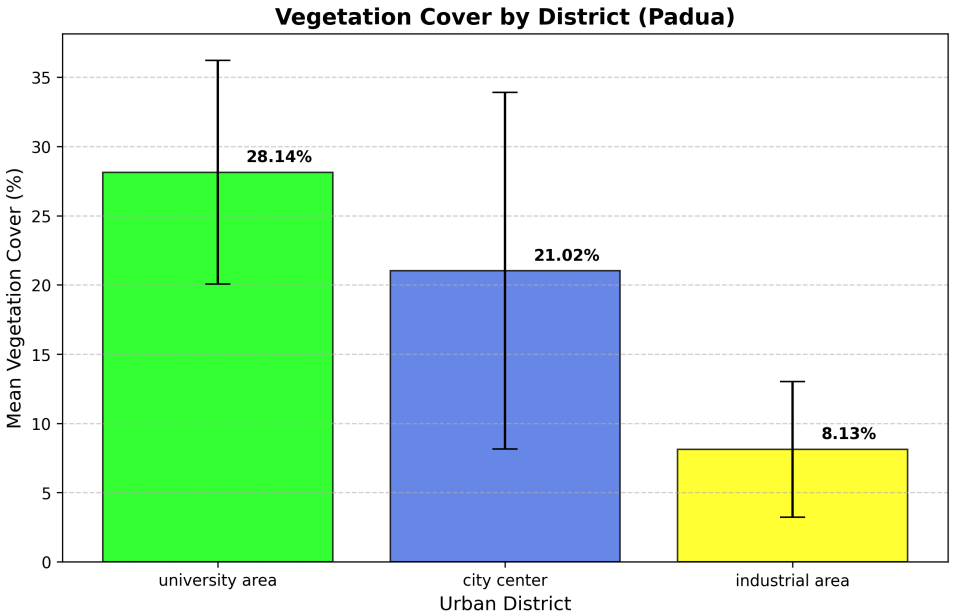


Table 2: Graphic data visualization.

3.1 Data Interpretation

The results highlight a strong correlation between urban land use and environmental quality:

- **University Area (28.14%):** This district emerged as the greenest zone. The high density of vegetation is attributed to the presence of gardens and green spaces situated along the Piovego river.
- **City Center (21.02%):** Despite the high density of historic buildings, the center maintains a significant amount of vegetation.
- **Industrial Area (8.13%):** This district recorded the lowest vegetation cover. The results reflect a landscape dominated by large-scale gray infrastructure (warehouses, logistics hubs and wide asphalt roads), where biological elements are often relegated to small, ornamental roadside strips or remain entirely absent.

The low standard deviation in the Industrial Area suggest a consistent low presence of greenery across the entire zone, whereas the high variability in the City Center suggests a heterogeneous green landscape.

4 Discussion

The analysis provides a quantitative overview of Padova's urban green distribution, yet the transition from satellite imagery to biological metrics involves technical challenges that must be addressed to contextualize the accuracy of our results.

4.1 Methodological Strengths and Limitations

The use of supervised machine learning method, such as Ilastik, represents a significant strength compared to simple color thresholding. By training the software to recognize texture and edges, we were able to distinguish between dark asphalt and dark green foliage which share similar spectral intensities. The consistency and performance of the classifier were evidenced by the training progression: while the initial images required extensive manual labeling to define the classes, subsequent images required only minimal refinement for minor details. This rapid reduction in manual intervention demonstrates that Ilastik successfully learned the features of the dataset, confirming its suitability for analyzing the urban landscape. However, several limitations and sources of error were identified during the classification phase:

- **Spectral Ambiguity:** The main challenge was the resolution of the images. It was noticed that in certain areas, even a human observer struggled to differentiate between vegetation and anthropic surfaces, leading to misclassification errors. This could be resolved in future studies by utilizing images with a higher zoom level or with a higher spatial resolution. Figure 8a illustrates a case where it is difficult to distinguish between a low-rise building with a mossy roof and a garden.

- **Biological Criteria:** A conscious decision was made to exclude arid patches of land from the vegetation count. This approach was based on the biological idea of measuring active green cover, capable of oxygen production and transpiration, over general permeable surfaces. Given that the dataset was taken during the peak growing season and the considered areas were not under cultivation, any arid patches was labeled as anthropic area. While this provides a more accurate map of the environmental health, it may results in lower percentages for poorly irrigated zones compared to the total permeable surface analysis. Figure 8b illustrates a representative example of an arid patch within the study area.

- **Shadow Dynamics:** Shadows introduced the greatest uncertainty. During the analysis it was observed that a tree casting a shadow on a lawn produces pixels very similar to a tree casting a shadow on a road. Without spatial context, the classifier occasionally mislabeled "anthropic shadows" as "vegetation" or vice versa. For instance, it was particularly hard to decipher in the City Center where narrow streets create complex shadow patterns. Furthermore, while the June 2025 acquisition maximized the visibility of tree canopies, it also maximized the area covered by shadows, potentially masking smaller shrubs or grass beneath the canopy layer. Figure 8c illustrates how shadows can obscure the surface, preventing reliable classification.

- **Artifacts:** The presence of moss on rooftops was a significant challenge. Because the pixel classifier only considers a spherical neighbourhood around each pixel (typically with a radius of 1 to 35 pixels), it lacks the broader spatial context necessary

to identify large-scale object. From a restricted perspective, the spectral signature of moss-covered tiles is nearly indistinguishable from low-lying vegetation. Consequently, this spectral similarity led to a slightly overestimation of the vegetation cover. Figure 8d illustrates an area where it is difficult to differentiate between rooftop moss and ornamental trees located on a terrace.



(a) Spectral ambiguity



(b) Biological criteria



(c) Shadow dynamics



(d) Artifacts

Figure 8: Examples of challenging visual scenarios for pixel classification.

4.2 Improvements

To create a dataset that is more representative of reality, several improvements are proposed:

- **Implementation of Autocontext:** As noted previously, while standard classification was effective, an Autocontext algorithm would be superior. By incorporating spatial context, the classifier can more accurately interpret local features, for instance, recognizing that a dark pixel surrounded by concrete is likely a shadow, whereas one surrounded by green is likely part of a tree. A similar, though less sophisticated, effect can be achieved in pixel classification by adjusting the Sigma scale. Increasing this parameter allows the software to aggregate information from a broader area, allowing it to extract contextual patterns from the wider neighborhood to improve classification accuracy.

- **Dataset Expansion:** Increasing the number of samples per district would lead to a more robust and reliable dataset, reducing the impact of local anomalies.

5 Conclusions

This analysis successfully quantified the vegetation cover across three key districts in Padova, revealing a significant disparity in green space distribution. The University Area emerged as the most biologically dense zone (28.14%), followed by the City Center (21.02%), while the Industrial Area showed a critical lack of vegetation, covering only (8.13%) of the sampled surface. From a methodological point of view, the combination of Ilastik's machine learning and ImageJ's automated quantification proved to be a reliable and reproducible pipeline; although challenges such as urban shadows and moss-covered rooftops introduced minor uncertainties. Future studies could further enhance this accuracy by implementing Autocontext workflows to better resolve complex urban textures.